Introduction to the Laws of Motion

Lesson 4: Force and Interaction

Force and Interaction – Study Notes

Key Concepts

1. **Force** – a push or pull that can change an object’s motion or shape.

2. **Contact Forces** – forces that act when objects are physically touching (e.g., friction, tension, normal force).

3. **Field (Non Contact) Forces** – forces that act at a distance through a field (e.g., gravity, electric, magnetic).

4. **Newton (N)** – the SI unit of force; 1 /N = 1 /kg·m /s {^{2}.

5. **Net Force & Motion** – the vector sum of all forces determines an object’s acceleration (Newton’s /1st /& /2nd laws).

Summary

A **force** is any interaction that can cause a change in an object’s velocity (speed or direction) or shape. Forces are vectors, meaning they have both magnitude and direction. They fall into two broad categories. **Contact forces** arise from physical contact between objects; examples include the normal force exerted by a surface, friction that opposes sliding, tension in a rope, and the force you feel when you push a door. **Field forces**, also called non contact forces, act over a distance without direct contact. Gravity pulls masses toward each other, while electric and magnetic forces act on charged particles and magnetic materials, respectively.

The standard unit for measuring force in the International System of Units (SI) is the **newton (N)**. One newton is the amount of force required to accelerate a 1 kilogram mass by 1 /meter per second squared (1 /N /= /1 /kg·m /s {^{2}). Understanding how forces combine (vector addition) to give a **net force** is essential, because the net force determines the resulting acceleration of the object according to Newton’s second law, **F = ma**.

Important Points

• **Force = mass × acceleration** (Newton’s 2nd law, **F = ma**).

• Forces are **vectors**; use arrows to show direction and add them tip to tail.

• **Contact forces** include:

- **Normal force (N)** – perpendicular to a surface.
- **Frictional force (f)** – parallel to a surface, opposes motion.
- **Tension (T)** – pulls along a rope or cable.
- **Applied force** – any push or pull you deliberately exert.

• **Field forces** include:

- **Gravitational force (Fg = mg)** – acts toward the Earth’s centre (or any massive body).

- **Electric force** – acts between charged objects.
- **Magnetic force** – acts on moving charges or magnetic materials.
  • **Newton (N)** definition: 1 /N = 1 /kg·m /s {².
  • **Net force** = vector sum of all individual forces; if net force = 0, the object is in equilibrium (constant velocity or at rest).
  • **Free body diagram**: a sketch that isolates an object and shows all forces acting on it; essential for solving problems.

Examples

Example 1 – Pushing a Box on a Floor

A 10 /kg wooden crate is pushed horizontally with a steady force of 50 /N. The coefficient of kinetic friction between the crate and the floor is 0.2.

1. **Identify forces:**
   - Applied force **F • = 50 /N** (horizontal, forward).
   - Weight **W = mg = 10 /kg × 9.8 /m /s {² = 98 /N** (downward).
   - Normal force **N** balances weight, so **N = 98 /N** (upward).
   - Friction **f – = ;Â dâ Ò ã" r “, ôâ Ò ’ãb ôâ¢¢ ††÷ izontal, opposite motion).
2. **Net horizontal force:**
   \(F_{\text{net}} = F • - f – = 50 /N - 19.6 /N = 30.4 /N\).
3. **Resulting acceleration:**
   \(a = F_{\text{net}} / m = 30.4 /N / 10 /kg = 3.04 /m /s {²\).

The crate accelerates forward at **3.0 /m /s {²**.

Example 2 – Object in Free Fall

A 2 /kg ball is dropped from rest near Earth’s surface. Ignore air resistance.

  • Only force: **gravity** \(F_g = mg = 2 /kg × 9.8 /m /s {² = 19.6 /N\) downward.
  • Net force = 19.6 /N !’ acceleration = **g = 9.8 /m /s {²** downward.

After 1 /s, its speed \(v = gt = 9.8 /m /s {² × 1 /s = 9.8 /m /s {¹\).

These examples illustrate how to list forces, draw a free body diagram, and apply **F = ma** to find acceleration or motion.

Quick Review

1. What is the SI unit of force and how is it defined?
2. Name two contact forces and two field forces.
3. A 5 /kg object experiences a net horizontal force of 20 /N. What is its acceleration?
4. Why does the normal force on a stationary object on a horizontal table equal its weight?
5. Sketch a simple free body diagram for a hanging mass suspended by a rope.

Further Reading

  • **Newton’s Three Laws of Motion** – deeper look at inertia, action–reaction, and dynamics.
  • **Friction** – static vs. kinetic friction, coefficients, and real world applications.

• **Gravitational Fields** – universal law of gravitation, weight vs. mass.
• **Vectors in Physics** – addition, resolution into components, and graphical methods.
• **Work, Energy, and Power** – how forces do work and change energy.